

EDUCATION

University of New South Wales

Master of Commerce in Business Analytics

Sydney, AU

Feb. 2026 – Dec. 2026 (Expected)

Hong Kong Baptist University

Bachelor of Business Administration (Honours) in Accounting

Hong Kong SAR

Sep. 2021 – Jun. 2025

- Studied at Beijing Normal-Hong Kong Baptist University (BNBU), Zhuhai Campus
- Second Class Honours

RESEARCH INTERESTS

- LLM evaluation
- Retrieval-augmented generation
- Knowledge graphs
- ESG and sustainability disclosure
- AI in accounting and finance

PUBLICATIONS & RESEARCH PAPERS

Ding, Y., Yang, Z., Lu, Y., Lan, L., Tang, Y., Yang, Z., Tang, X., **Hu, J.**, Ju, M., Zhang, W., Yang, W., Xu, X., & Zhang, W. (2026). EulerESG: A unified platform for standards-aware ESG disclosure analysis. In *Proceedings of the IEEE International Conference on Data Mining (ICDM), Demo Track*, Shenyang, China. (Accepted).

Hu, J., & Wang, Y. (2025). *The Impact of Corporate Environmental, Social and Governance (ESG) Disclosure on Corporate Investment Efficiency* [Unpublished undergraduate final year project]. Beijing Normal-Hong Kong Baptist University. [[PDF](#)] [[Code](#)]

RESEARCH EXPERIENCE

Research Assistant – Remote)

Sep. 2026 – Present

Vecton AI Pty Ltd

Sydney, AU

Direct Supervisor: Bocheng Han

- Undertake technical training in retrieval-augmented generation and LLM agents, covering query rewriting, hybrid retrieval, reranking, evaluation, tool use and agent memory.
- Study Transformer fundamentals, including tokenization, attention, sampling and training, alongside introductory work with FastAPI, LangChain and vector databases.

Student Researcher – Remote

May. 2026 – Present

RAIDS Lab, University of Sydney

Sydney, AU

Supervisor: Dr. Zhengyi Yang [[Details](#)]

- Conduct research on ESG disclosure analysis, focusing on the reliability and evidence grounding of assessments generated by large language models.
- Contributed to manuscript revision, experiments and system review for EulerESG, including suggestions for improving the front-end application. Co-authored the paper accepted to the IEEE ICDM 2026 Demo Track.
- Collaborate with Zherui Wang on ESG data diagnostics and benchmark design and develop an ontology-guided research proposal addressing evidence grounding and cross-dimension bias in LLM-generated assessments.

SELECTED PROJECTS

Reproducible Diagnostics and Benchmark Design for ESG Disclosure Assessment

Supervisor: Dr. Zhengyi Yang

Collaborator: Zherui Wang, UNSW CSE

UNSW, Sydney

Aug. 2026 – Present

- Developed reproducible diagnostics for 25,224 ESG disclosure records covering 37 ASX-listed entities and 53 report-years under AASB S1/S2, GRI and SASB, comparing model outputs with human-reviewed annotations.
- Found that a simple evidence-presence rule matched 99.92% of existing disclosed/not-disclosed labels, motivating evaluation against individual disclosure requirements rather than reliance on aggregate labels.
- Documented data-quality issues and selected a benchmark subset based on label balance. Designed a blind double-annotation protocol for 300 items, including annotator calibration, agreement assessment and adjudication.

Carbon Chain – Cross-Border Sustainability Compliance Verification

Supervisor: Dr. Zhengyi Yang

China International College Students' Innovation Competition 2027

Jun. 2026 – Present

Project for the China International College Students' Innovation Competition 2027

- Lead a team working on backend development, frontend development and evaluation to adapt EulerESG to Chinese disclosure requirements.
- Designed and implemented a four-layer graph linking ESG entities, claims and evidence, provenance, and temporal information to assess support for disclosure claims while distinguishing independent evidence sources.
- Collected 48 reports from 10 A-/H-share issuers and piloted the workflow on CATL's carbon-neutrality disclosures, identifying restatements and differences in reporting standards as obstacles to comparison.

ESG-Augmented Corporate Credit Risk Prediction and Explainability Analysis

2026, UNSW, Sydney [\[Code\]](#)

- Built an XGBoost pipeline using financial, market and ESG features from over 42,000 firm-year observations of Chinese A-share companies. Achieved $R^2 = 0.68$ for KMV distance-to-default prediction and ROC-AUC = 0.92 for next-year ST/*ST classification.
- Evaluated ESG's incremental predictive value using firm-grouped cross-validation and ablation analysis, finding a 1.25% reduction in RMSE ($p = 0.003$). Tested robustness across alternative risk measures and used SHAP to interpret model predictions.

Multi-Factor Quantitative Stock Selection & Backtesting

2026, UNSW, Sydney [\[Code\]](#)

- Built a factor-research pipeline covering approximately 499,000 stock-month observations across 5,546 A-share firms, including delisted companies. Constructed eight factors with winsorization, industry and size neutralization, and reporting lags to reduce the risk of look-ahead bias.
- Evaluated factors using rank IC, ICIR and decay analysis. A walk-forward LightGBM composite achieved an ICIR of 0.72 and a gross long-short quintile Sharpe ratio of 0.93, before transaction costs and practical short-selling constraints.

The Impact of Corporate ESG Disclosure on Corporate Investment Efficiency

Final Year Project submitted to BNBU [\[PDF\]](#) [\[Code\]](#)

Supervisor: Assoc. Prof. Man Hung Alvin Cheng [\[Details\]](#)

Feb. 2024 – Apr. 2025

- Examined the relationship between ESG disclosure, information transparency and investment efficiency using 19,103 firm-year observations of Chinese A-share companies.
- Constructed alternative investment-efficiency measures and applied panel regressions, instrumental-variable estimation, system GMM and bootstrap mediation analysis, comparing findings across specifications and documenting differences in statistical significance.

Chinese White Dolphin Habitat Loss and Conservation in Western Hong Kong

2023, BNBU [\[Code\]](#)

- Built a reproducible PDF-to-panel pipeline (PyMuPDF, pandas) parsing ten years of AFCD marine-mammal monitoring reports into a validated area $\times$ monitoring-period panel of 2,209 sightings and 7,023 survey-effort records, aligned with EPD station-level water-quality data.
- Applied a Negative Binomial GLM with area fixed effects and a survey-effort offset, kernel utilization distributions, a Poisson tensor-smooth spatial GAM, and Pettitt / Mann-Kendall change-point tests, estimating an effort-controlled decline of $\sim 12.4\%$ per year (IRR = 0.876) and a 50% core-use area contracting from 45 to 26 km² with a 6.5 km south-westward shift.

INDUSTRY EXPERIENCE

Business Analyst

Vecton AI Pty Ltd

Sep. 2026 – Sep. 2026

Sydney, AU

Financial planning and funding analysis for a project in the China International College Students' Innovation Competition 2026

Financial Analysis Intern

Jinchuan Group Corporation Limited

Jun. 2024 – Sep. 2024

Jinchang, CN

Capital Budgeting Intern

State Grid Corporation of China

Jun. 2022 – Sep. 2022

Jinchang, CN

Investment Research Intern

Gansu Electric Power Investment Group Corporation Limited

Jun. 2021 – Sep. 2021

Lanzhou, CN

HONORS & AWARDS

Dean’s List	BNBU, 2022-2023, 2023-2024, 2024-2025
Bronze Award of Servant Leadership	Student Affairs Office, BNBU, 2024
Excellent Student of Environmental Awareness	Whole Person Education, BNBU, 2023

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Stata, MATLAB, SAS, SPSS, Excel VBA
- **Research software engineering:** pytest, deterministic seeding, provenance tracking, reproducibility verification, PDF parsing and OCR-assisted extraction, Git; RDF/OWL, SHACL, PROV-O, knowledge graphs, RAG, LLM-as-judge evaluation
- **Machine Learning and statistics:** PyTorch, scikit-learn, XGBoost, SHAP; panel fixed effects, 2SLS, system GMM, bootstrap inference
- **Research Data:** CSMAR, Wind, Bloomberg, WRDS, TuShare, SSE/SZSE/HKEX filings, ESG and sustainability reports
- **Languages:** Mandarin Chinese (native), English (proficient)